Deep Learning I — Final Project Report

Course: Deep Learning I: Neural Networks (Semester 4) · Section A · Group 1

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Abstract

We present two studies. **Task 1 (applied)** forecasts air temperature **12 hours ahead** from the Jena Climate dataset, comparing a moving-average/persistence baseline, a vanilla RNN, and an LSTM, and analysing the effect of the lookback-window length. **Task 2 (experimental)** trains a small VGG-style CNN on CIFAR-10 (82.2% test accuracy) and visualises its first-layer filters and feature maps at an early vs. a late layer to show how representations grow more abstract with depth.

Task 1 — Time Series Forecasting (Weather)

1.1 Problem & data

Predict the temperature (`T (degC)`) **12 hours ahead** from the Jena Climate dataset (2009–2016, 10-min readings resampled to hourly → ~70k points; ~49k supervised windows). Strong daily seasonality is visible in the one-week zoom (Fig. 1). Data is split **chronologically** 70/15/15 and standardised using **train** statistics only (no leakage). Inputs are sliding windows of `LOOKBACK` hours → the value 12 h after the window.

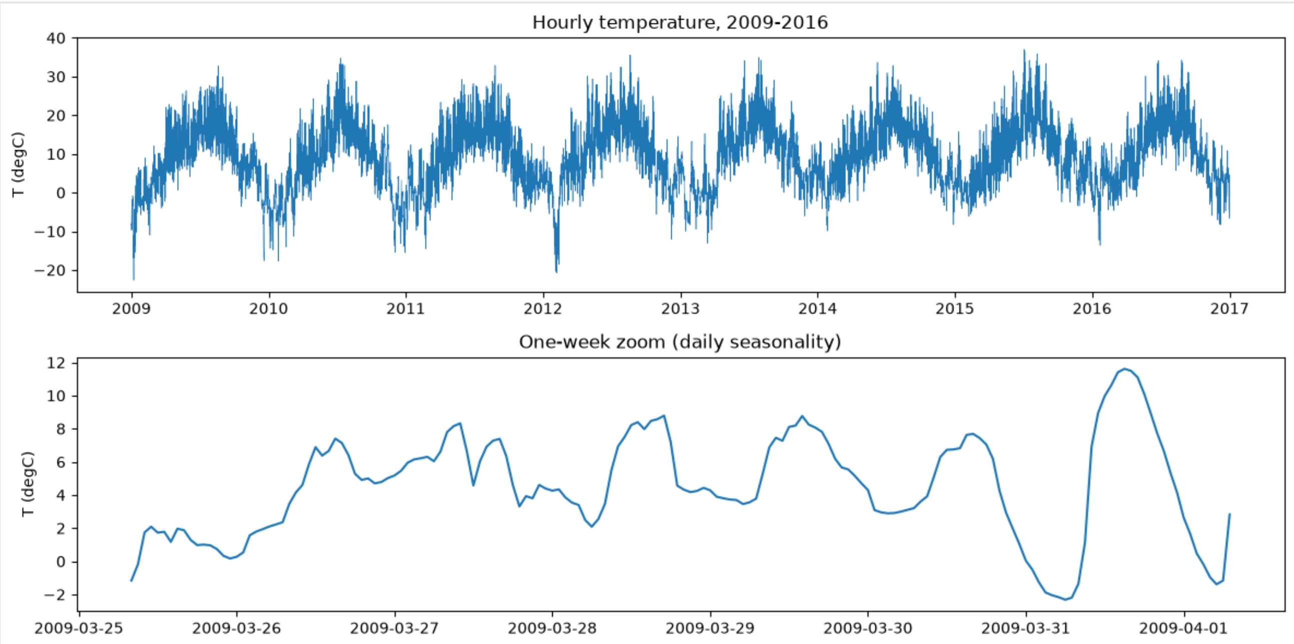

Fig. 1 — Full hourly series (2009–2016) and a one-week zoom showing the daily cycle.

1.2 Methods

- **Baseline** — persistence (forecast = last observed value) and moving average of the window. No training.
- **Vanilla RNN** — `nn.RNN`, hidden=64, last hidden state → linear head.
- **LSTM** — `nn.LSTM`, hidden=64, same head. Identical training (Adam, MSE, 12 epochs) so the only difference is the recurrent cell. All metrics (MAE, RMSE) are reported in **°C** after de-normalisation.

1.3 Results

Model	MAE (°C)	RMSE (°C)
LSTM	**2.19**	**2.81**
Vanilla RNN	2.22	2.86
Moving average	2.96	3.73
Persistence	4.18	5.41

The **LSTM is best**, cutting RMSE by **~48%** versus the persistence baseline (2.81 vs 5.41). At a 12-hour horizon persistence is badly out of phase — it predicts the temperature from 12 h earlier, i.e. roughly the opposite point of the day/night cycle (Fig. 3, light blue), which is exactly why the recurrent models win so clearly here.

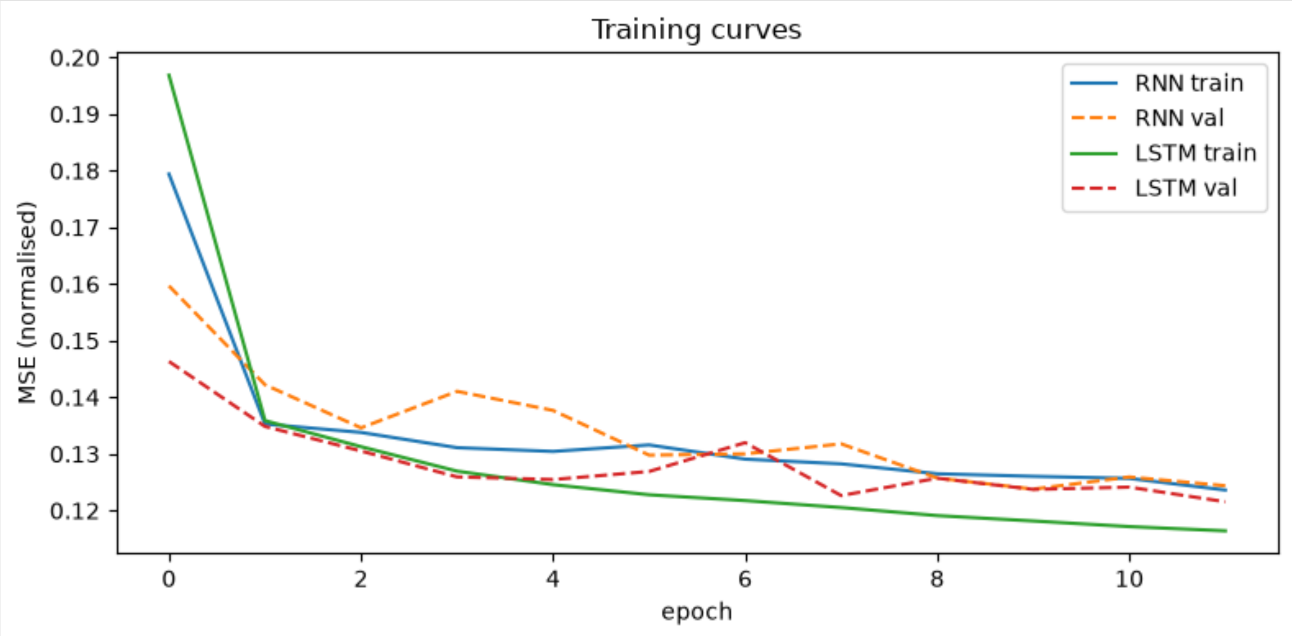

Fig. 2 — Training/validation MSE (normalised). Both converge; the LSTM reaches a lower val loss.

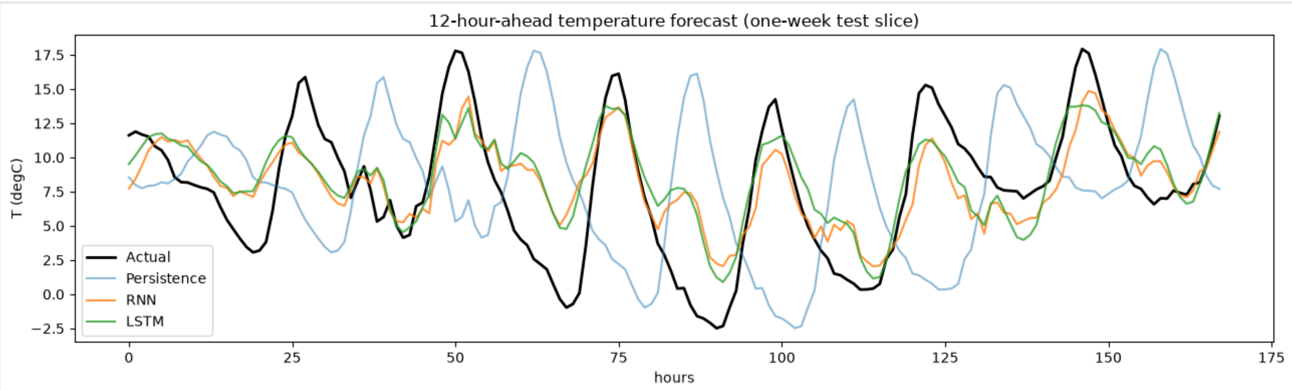

Fig. 3 — 12-h-ahead forecast over a one-week test slice. RNN/LSTM track the daily cycle; persistence is a full half-day out of phase.

1.4 Ablation — lookback window length

We retrain RNN and LSTM for lookbacks `[6, 12, 24, 48, 96]` h (Fig. 4).

Lookback (h)	RNN RMSE	LSTM RMSE
6	3.92	4.16
12	3.06	2.99
24	2.96	2.86

Lookback (h)	RNN RMSE	LSTM RMSE
48	2.88	2.82
96	2.88	**2.74**

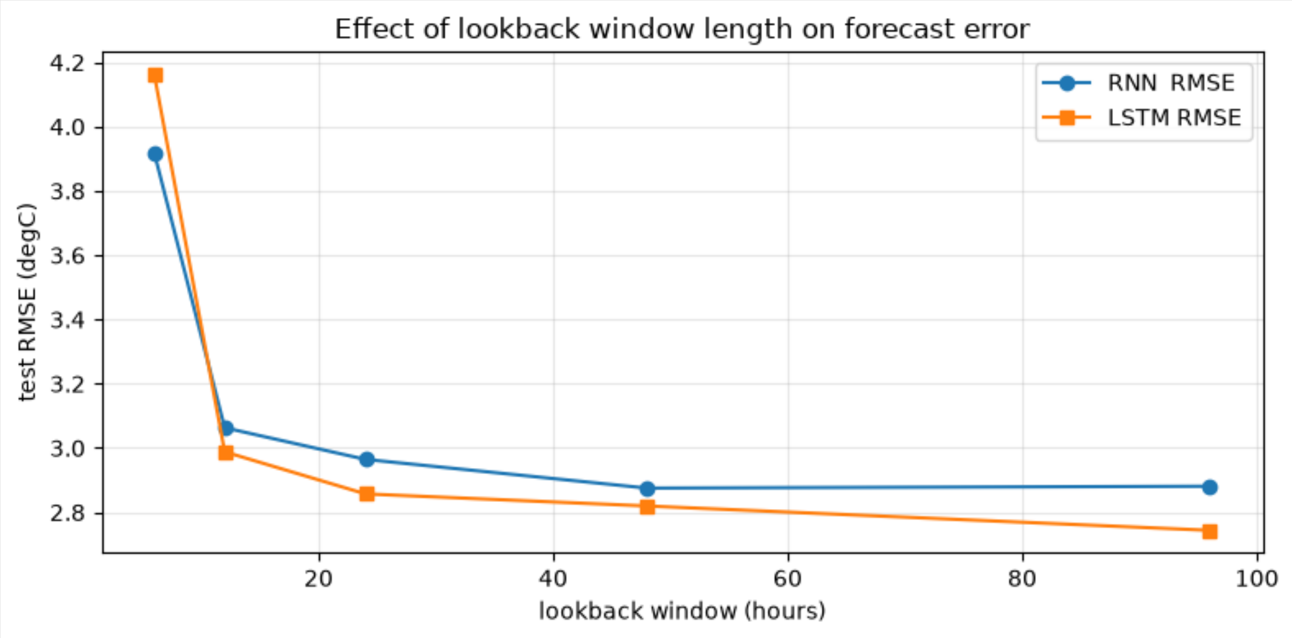

Fig. 4 — Test RMSE vs lookback window length.

Analysis. A 6-hour window is **too short** — it can't see the daily cycle, and error is worst (≈3.9–4.2 °C). Error drops sharply as the window approaches ~24 h (one full day), then **flattens**. Beyond that the two cells diverge: the **LSTM keeps improving** out to 96 h (2.82 → 2.74) because its gates retain useful long-range context, whereas the **vanilla RNN plateaus and slightly worsens** (2.875 → 2.881), consistent with its weaker long-sequence memory. Practical takeaway: ~24–48 h is the sweet spot for the RNN; the LSTM benefits from longer windows.

Task 2 — CNN Filter Visualization (CIFAR-10)

2.1 Problem & model

Train a 4-conv VGG-style CNN (conv1 32 → conv2 64 → conv3 128 → conv4 128, two max-pools, FC head; **2,340,810 params**) on CIFAR-10 with light augmentation (random crop + flip) and channel normalisation.

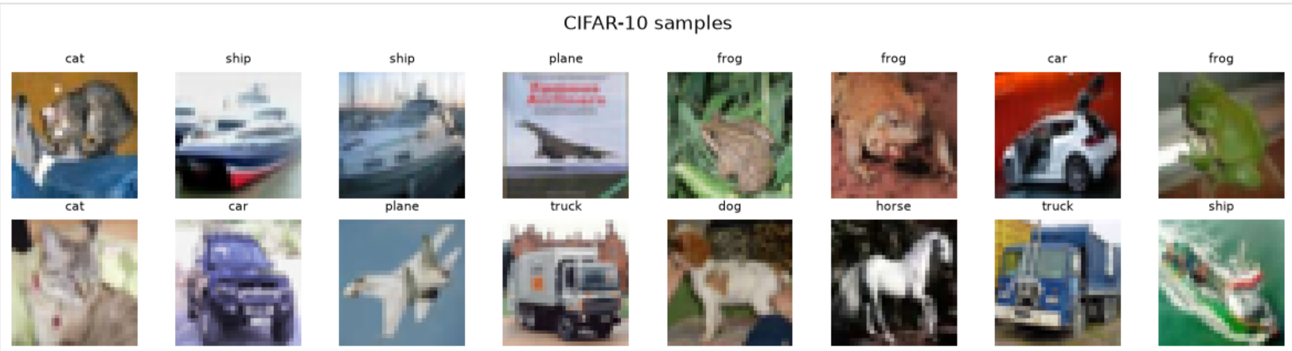

Fig. 5 — Sample CIFAR-10 images.

2.2 Results

Final test accuracy: 82.2% (12 epochs). Curves are healthy with a small train/test gap.

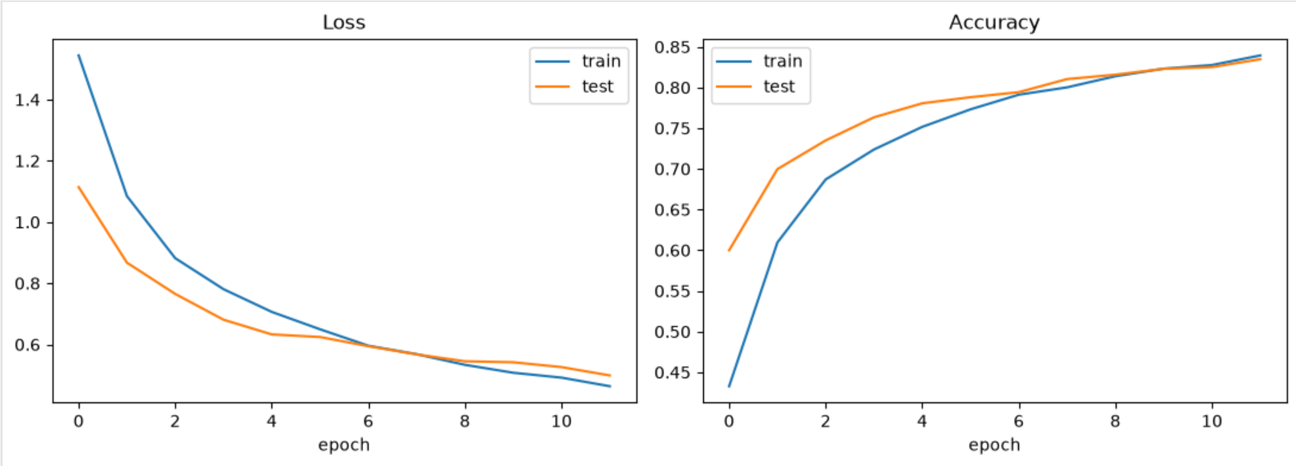

Fig. 6 — Training/test loss and accuracy.

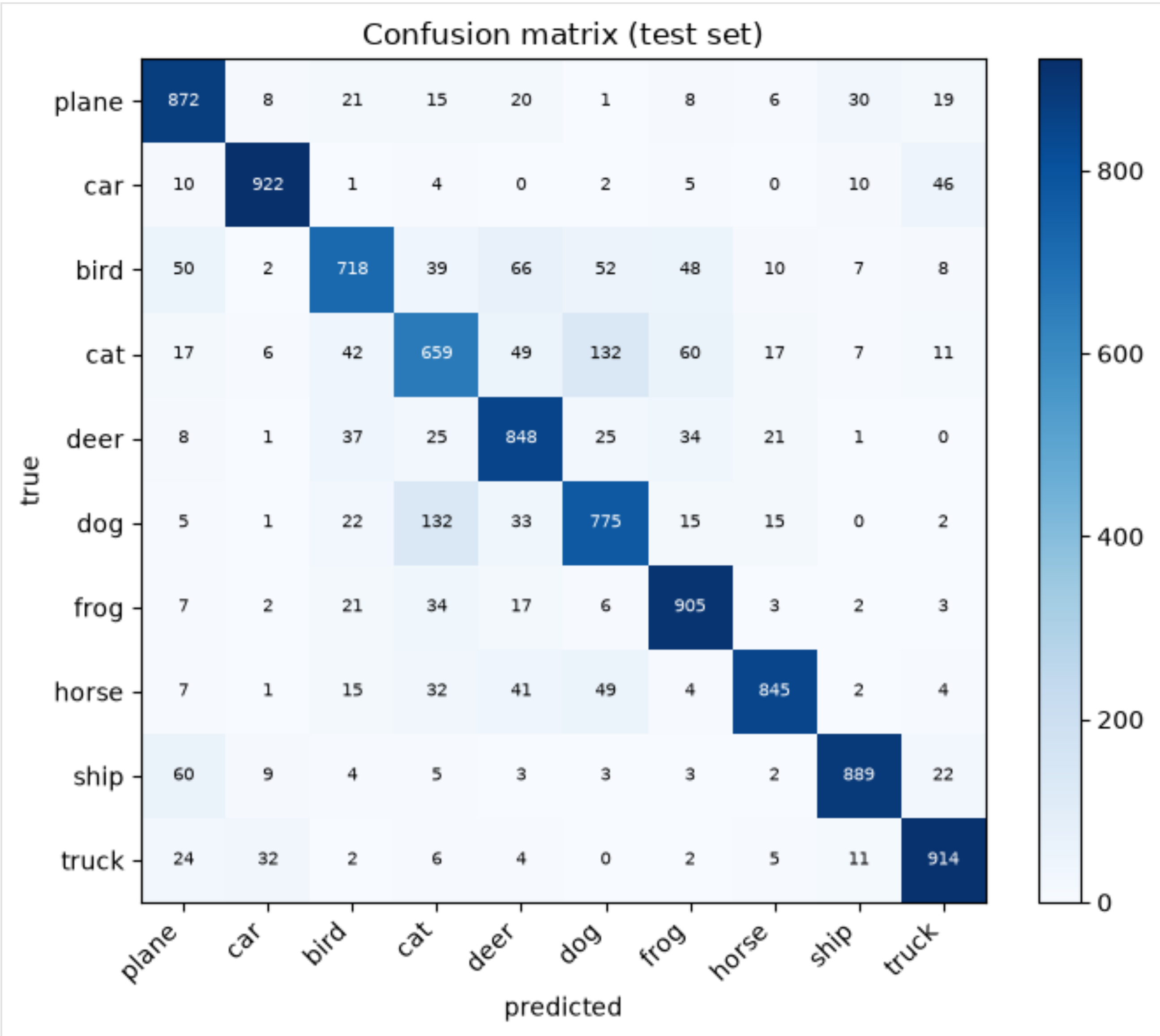

Fig. 7 — Test-set confusion matrix.

Per-class accuracy is highest for visually distinct classes (ship 92%, plane 90%, truck 89%, deer 87%) and lowest for visually similar animals (**cat 61%**, **bird 70%**, dog 75%). Most errors fall between semantically close classes — notably **cat↔dog** (128 + 92 misclassified) and **car↔truck** (77) — which is the expected CIFAR-10 failure mode.

2.3 Filter & feature-map visualization

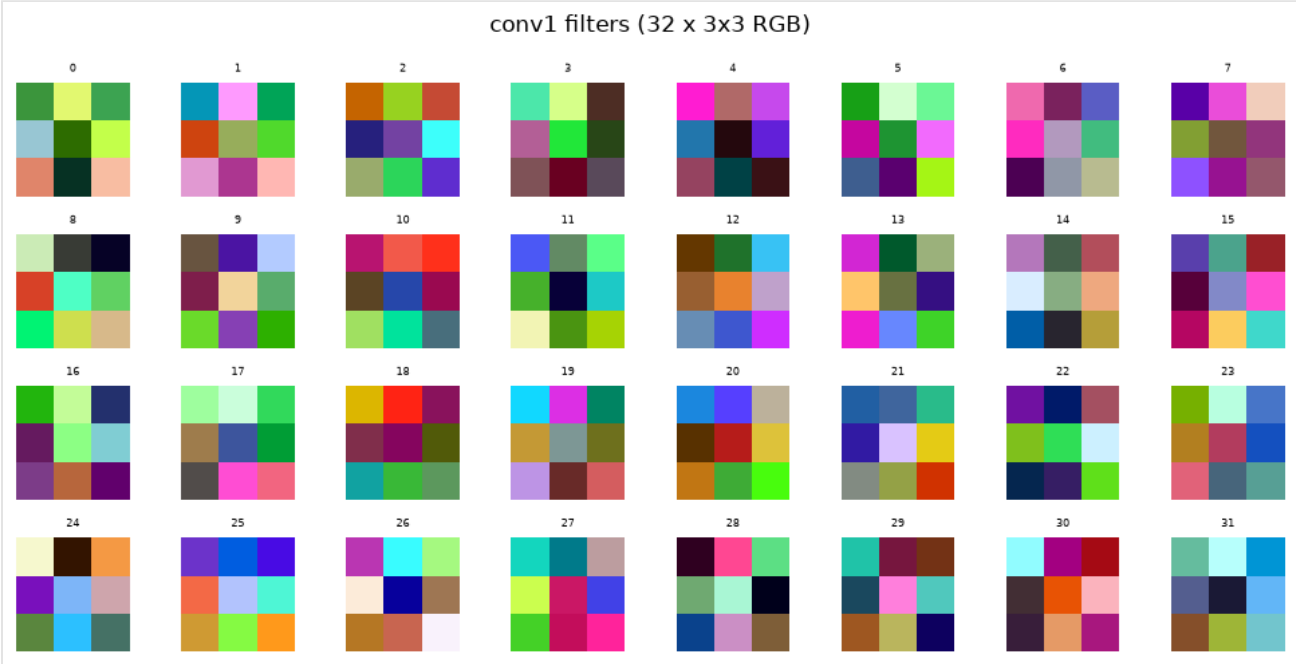

Fig. 8 — The 32 first-layer (conv1) filters: low-level edge / colour-opponent detectors.

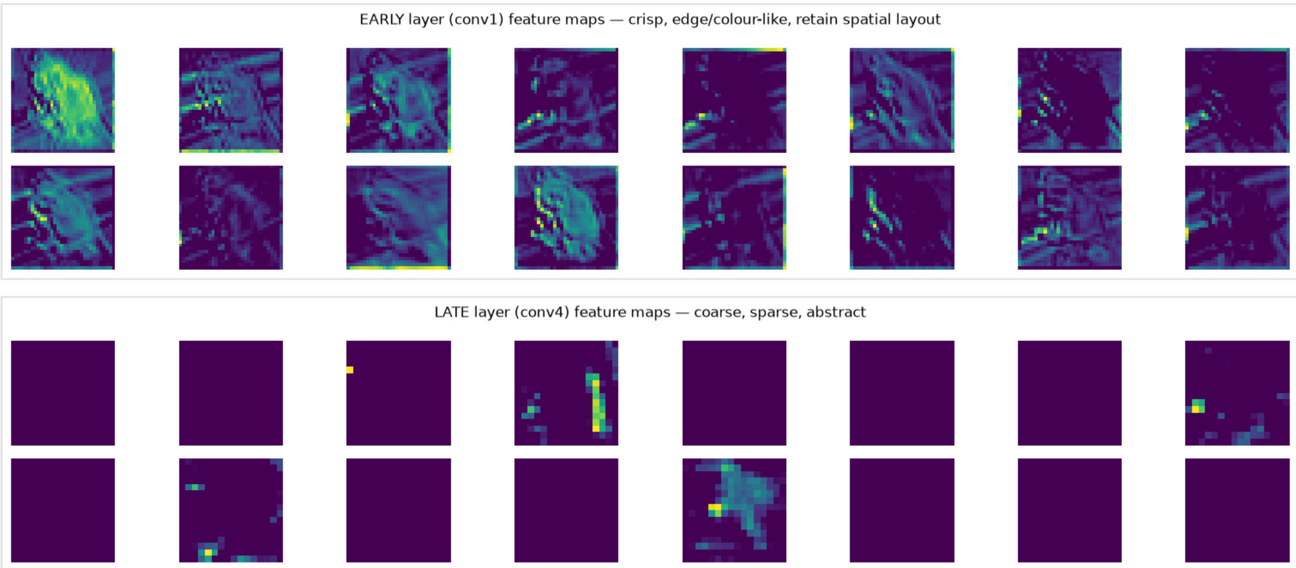

Fig. 9 — Feature maps for one input. Early (conv1, 32×32) are sharp and retain the object outline; late (conv4, 16×16) are coarse, sparse, and abstract.

2.4 Analysis — abstraction with depth

The network builds a hierarchy: pixels → edges/colours (conv1) → textures/parts → abstract, class-discriminative features (conv4). Early maps answer *where* (spatial detail); deep maps answer *what* (semantics), trading resolution for meaning. Many conv4 channels are silent for a given image (sparse, selective activations) — a hallmark of learned high-level detectors. This hierarchy is *why* deep stacks of small convolutions classify well, and it is consistent with the confusion-matrix errors clustering among visually similar classes.

Conclusion

For 12-h-ahead temperature forecasting, **LSTM > RNN ≫ moving average ≫ persistence**, and the lookback sweep shows error falling until ~one daily period then flattening, with the LSTM uniquely benefiting from longer context. The CNN study makes the depth→abstraction story concrete and visual at 82.2% accuracy.

Limitations / future work: multi-step (sequence) forecasting and exogenous features (pressure, humidity) for Task 1; deeper nets / Grad-CAM for Task 2.

Member contributions

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Shreyansh Arora	24BCS10252	shreyansh.24bcs10252@sst.scaler.com	Project coordination; Task 1 data pipeline & preprocessing
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Mohammed Rehan	24BCS10620	mohammed.24bcs10620@sst.scaler.com	Task 1 baselines, metrics (MAE/RMSE) & evaluation
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Shubh Shukla	24BCS10093	shubh.24bcs10093@sst.scaler.com	Report writing & slide deck preparation

> *Note: every member ran both notebooks end-to-end and can explain any part for the viva.*

References

- Jena Climate dataset (Max Planck Institute for Biogeochemistry).
- Krizhevsky, *Learning Multiple Layers of Features from Tiny Images* (CIFAR-10), 2009.
- Hochreiter & Schmidhuber, *Long Short-Term Memory*, Neural Computation, 1997.